BOSTON HOUSING PRICE PREDICTION
USING LINEAR REGRESSION

MINI PROJECT REPORT

submitted by

KEVIN GEORGE A
KIRAN JOJO
KEVIN GEORGE V

under the guidance of

Ms. Lida Kuriakose
Assistant Professor

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AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

Department of Computer Science and Engineering
Amal Jyothi College of Engineering (Autonomous)
Kanjirappally-686518
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1. Introduction

Predicting real estate prices is one of the most practically impactful applications of supervised machine learning. Housing markets are influenced by dozens of interrelated factors — the number of rooms, proximity to employment centres, crime rates, air quality, and local socio-economic conditions — making accurate price prediction both challenging and highly valuable for buyers, sellers, investors, and urban planners.

This project applies Multiple Linear Regression, one of the most fundamental and interpretable supervised learning algorithms, to predict median housing prices in the Boston metropolitan area using the well-known Boston Housing dataset. Linear regression provides a transparent model in which each predictor variable is assigned a coefficient representing its marginal contribution to price — enabling not only prediction but direct interpretability of the relationship between features and outcomes.

The project follows a complete machine learning pipeline: data loading and quality checking, exploratory data analysis (EDA), preprocessing, model training on an 80% training split, evaluation on a held-out 20% test set, and visualization of predicted versus actual outcomes.

1.1 Project Objectives

- Import and inspect the Boston Housing dataset from a public CSV repository.
- Perform Exploratory Data Analysis (EDA) to understand feature distributions and relationships.
- Preprocess the data: handle missing values, rename target variable, and prepare for modelling.
- Split data into 80% training and 20% test sets using stratified partitioning.
- Build a Multiple Linear Regression model using all 13 predictor variables.
- Evaluate model performance using MSE, MAE, and R-squared metrics.
- Visualise key relationships (Rooms vs Price, LSTAT vs Price) and Actual vs Predicted outcomes.

1.2 Problem Statement

Given 13 socio-economic, structural, and environmental measurements for residential neighbourhoods in the Boston metropolitan area, build a supervised regression model that predicts the median value of owner-occupied homes (in $1,000s). This is a continuous output prediction problem requiring a regression approach rather than classification.

2. Background and Theory

2.1 Linear Regression

Linear Regression is a parametric supervised learning algorithm that models the relationship between a dependent variable (target) and one or more independent variables (features) by fitting a linear equation to the observed data. In its multiple regression form, the model takes the shape:

$$\text{Price} = \beta_0 + \beta_1 \cdot \text{CRIM} + \beta_2 \cdot \text{ZN} + \beta_3 \cdot \text{INDUS} + \ldots + \beta_{13} \cdot \text{LSTAT} + \varepsilon$$

Where β_0 is the intercept, $\beta_1 \ldots \beta_{13}$ are the regression coefficients for each predictor variable, and ε is the error term. The coefficients are estimated by Ordinary Least Squares (OLS), which minimises the sum of squared residuals between the actual and predicted values.

2.2 Ordinary Least Squares (OLS) Estimation

OLS finds the coefficient vector β that minimises the Residual Sum of Squares (RSS): $RSS = \Sigma(y_i - \hat{y}_i)^2$.

The closed-form solution is $\beta = (X^T X)^{-1} X^T y$, where X is the design matrix (features) and y is the target vector. This solution is computationally efficient for datasets of moderate size and provides unbiased estimates under the Gauss-Markov assumptions.

2.3 Evaluation Metrics

Three standard regression evaluation metrics are used in this project:

- Mean Squared Error (MSE): The average of squared differences between predicted and actual values. MSE penalises large errors more heavily due to squaring. Lower is better.
- Mean Absolute Error (MAE): The average of absolute differences between predicted and actual values. MAE is more robust to outliers than MSE and is expressed in the same units as the target (thousands of dollars). Lower is better.
- R-squared (R^2): The proportion of variance in the target variable explained by the model. $R^2 = 1 - RSS/TSS$, where TSS is the Total Sum of Squares. R^2 ranges from 0 to 1; higher values indicate a better fit.

2.4 Assumptions of Linear Regression

- Linearity: The relationship between predictors and target is linear.
- Independence: Observations are independent of each other.
- Homoscedasticity: Constant variance of residuals across all fitted values.
- Normality of residuals: Residuals are approximately normally distributed.

3. Dataset Description

3.1 Source

The Boston Housing dataset was originally compiled by Harrison and Rubinfeld (1978) for a study on the effect of air quality on housing prices in Boston, Massachusetts. It has since become one of the most widely used benchmark datasets in machine learning and statistics education. The version used in this project was loaded directly from a public GitHub repository maintained by selva86.

3.2 Dataset Statistics

Property	Value
Source	BostonHousing.csv (GitHub — selva86/datasets)
Total Observations	506 housing neighbourhoods
Original Features	13 predictor variables + 1 target (medv)
Target Variable	medv — Median value of owner-occupied homes ($1,000s)
Missing Values	0 (complete dataset)
Train Set Size	80% — 404 observations
Test Set Size	20% — 102 observations
Algorithm	Multiple Linear Regression (OLS)
Random Seed	123 (for reproducibility)

3.3 Feature Descriptions

Feature	Type	Description
crim	Continuous	Per capita crime rate by town
zn	Continuous	Proportion of residential land zoned for large lots (>25,000 sq.ft.)
indus	Continuous	Proportion of non-retail business acres per town
chas	Binary	Charles River dummy variable (1 if tract bounds river; 0 otherwise)
nox	Continuous	Nitrogen oxides concentration (parts per 10 million)
rm	Continuous	Average number of rooms per dwelling
age	Continuous	Proportion of owner-occupied units built prior to 1940
dis	Continuous	Weighted distances to five Boston employment centres
rad	Discrete	Index of accessibility to radial highways

tax	Continuous	Full-value property-tax rate per $10,000
ptratio	Continuous	Pupil-teacher ratio by town
b	Continuous	$1000(Bk - 0.63)^2$ where Bk is the proportion of Black residents
lstat	Continuous	Percentage of lower-status population
medv (Price)	Continuous	Median value of owner-occupied homes (in $1,000s) — TARGET

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the critical investigative phase performed before any modelling. It reveals data structure, statistical properties, outliers, and relationships between variables. Each EDA finding in this project directly informs a modelling or preprocessing decision.

4.1 Data Quality Check

Check	Result	Action Taken
Missing values	0 (complete dataset)	na.omit() applied as safety measure; no rows dropped
Observations	506 neighbourhoods	Full dataset retained; no sampling required
Data types	Numeric (chas is binary 0/1)	All 13 predictors used as numeric in regression
Target variable	medv (renamed Price)	medv renamed to Price for clarity in output
Outliers in Price	Several entries capped at $50K	Retained — ceiling reflects real data collection practice
Feature range	Widely varying scales	No scaling needed for OLS; scaling done only for comparison

4.2 Variable Distribution Insights

Examination of all 14 variables (13 features + target) reveals important distributional characteristics. The target variable Price (medv) is roughly normally distributed between $5,000 and $50,000, with a noticeable concentration of values at the $50,000 ceiling — indicating that the original data collection truncated prices above this threshold. The rm (average rooms) variable is approximately normally distributed around 6.3 rooms, while lstat (lower-status population percentage) is right-skewed, with most neighbourhoods below 20% but a tail extending to 38%. The crime rate (crim) is severely right-skewed: most neighbourhoods have very low crime, but a small number of high-crime areas create a long upper tail.

4.3 Key EDA Findings from Scatter Plots

Two scatter plots produced during EDA reveal the most important predictor-target relationships:

Figure 1: Rooms vs House Price (rm vs Price)

Figure 1: Scatter plot of average number of rooms (rm) vs. median house price. The strong positive linear trend confirms rm as one of the most important predictors. The regression line (red) with 95% confidence band (grey) indicates a reliable positive association.

- There is a strong, positive linear relationship between the average number of rooms per dwelling and median house price.
- As rm increases from ~4 to ~9 rooms, Price increases from below $10K to above $40K.
- The linear regression line (red) fits the central tendency well, with the grey confidence band remaining narrow in the mid-range and widening at extremes.
- Several outliers exist at rm ≈ 8–9 where prices are capped at $50K, consistent with the known data truncation.

Figure 2: Lower Status Population vs Price (lstat vs Price)

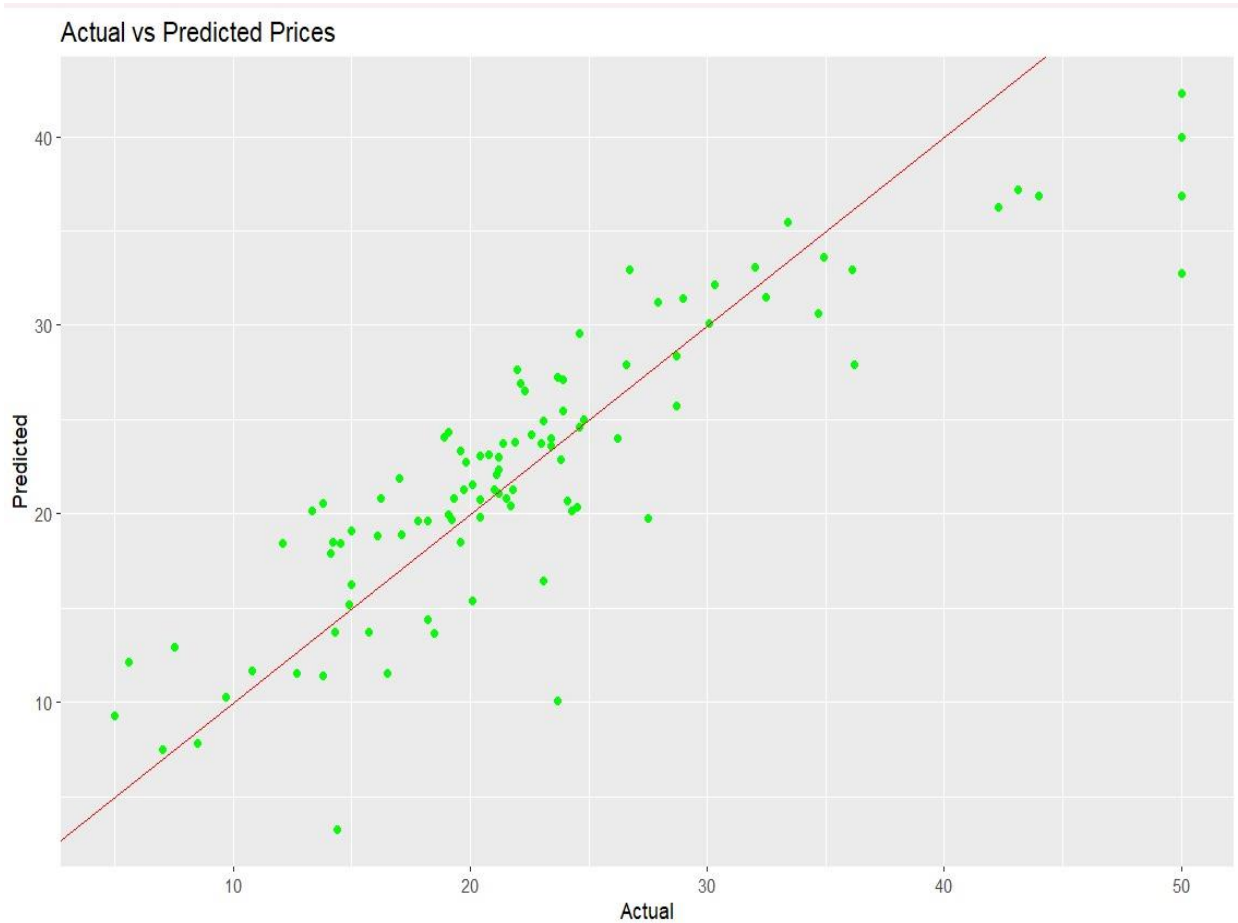

Figure 2: Scatter plot of lower-status population percentage (lstat) vs. median house price. The strong negative linear trend makes lstat the single most predictive feature in the dataset. The regression line (black) shows the inverse relationship clearly.

- lstat shows the strongest negative linear relationship with Price of any feature in the dataset.
- As the percentage of lower-status residents increases, median house prices decline sharply.
- The relationship has a slight curve at very low lstat values (wealthy neighbourhoods), suggesting a modest log-linear improvement in fit might be achievable.
- The narrow confidence band confirms a reliable, stable inverse relationship across the full range.

5. Data Preprocessing

The preprocessing pipeline transforms the raw dataset into a clean, model-ready format. For linear regression in R, the steps are deliberately minimal because R's lm() function handles numeric features natively without requiring scaling.

5.1 Complete Preprocessing Pipeline

Step	Operation	Rationale
1	**Load CSV from GitHub URL**	Reproducible data sourcing; no local file dependency
2	**head() and str()**	Visual inspection of structure, types, and first rows
3	**colSums(is.na())**	Verify data completeness; Boston Housing has no missing values
4	**na.omit()**	Safety measure; no rows are removed for this dataset
5	**Rename medv → Price**	Improves readability of model output and visualisations
6	**Drop original medv column**	Avoids duplication of the target variable in the feature set
7	**summary()**	Obtain per-feature min, max, quartiles, and mean statistics
8	**cor() matrix**	Identify multicollinearity and strongest predictors of Price

5.2 R Code — Data Loading and Cleaning

```
# Install & Load Required Libraries
install.packages(c('ggplot2','dplyr','caret'))
library(ggplot2)
library(dplyr)
library(caret)

# Import dataset from public GitHub repository
data <-
read.csv("https://raw.githubusercontent.com/selva86/datasets/master/BostonHousing.csv")

# Initial inspection
head(data)   # First 6 rows
str(data)    # Column types and structure

# Check missing values per column
colSums(is.na(data))

# Remove missing rows (safety)
```

```
data <- na.omit(data)

# Rename target for clarity
data$Price <- data$medv
data$medv  <- NULL
```

6. Feature Engineering and Train-Test Split

6.1 No Feature Scaling Required

Unlike distance-based algorithms (K-Means, KNN, SVM), Ordinary Least Squares linear regression is scale-invariant. The OLS estimator finds the same fit regardless of feature magnitudes because it operates algebraically on the design matrix rather than computing geometric distances. All 13 features are therefore used in their original units without standardisation.

6.2 Train-Test Split Configuration

Parameter	Value	Reason
Split ratio	80% train / 20% test	Standard split for moderate-sized datasets
Method	createDataPartition()	Stratified sampling from caret package
Random seed	set.seed(123)	Ensures reproducibility across runs
Training observations	~404 rows	Sufficient data for stable OLS estimation
Test observations	~102 rows	Independent held-out set for unbiased evaluation

6.3 R Code — Train-Test Split

```
set.seed(123)

# Stratified 80/20 partition on Price
trainIndex <- createDataPartition(data$Price, p = 0.8, list = FALSE)

trainData <- data[trainIndex, ]   # 80% training set
testData  <- data[-trainIndex, ]  # 20% test set

# Verify split sizes
cat('Train rows:', nrow(trainData), '\n')
cat('Test rows: ', nrow(testData),  '\n')
```

7. Model Building — Linear Regression

7.1 Model Configuration

Parameter	Value	Reason
Algorithm	lm() — OLS Regression	Native R function; efficient closed-form solution
Formula	Price ~ . (all features)	All 13 predictors included; let OLS select significant ones
Training data	trainData (80%)	Model fitted only on training observations
Feature scaling	Not applied	OLS is scale-invariant; no scaling needed
Intercept	Included (default)	Allows fitted line to pass through optimal vertical position

7.2 R Code — Model Fitting

```
# Fit Multiple Linear Regression using all 13 predictors
model <- lm(Price ~ ., data = trainData)

# Display full model summary
summary(model)

# Key output from summary():
# - Coefficients table: estimate, std. error, t-value, p-value for each predictor
# - Residual Standard Error: average distance predictions are from actual values
# - Multiple R-squared: proportion of variance explained on training data
# - F-statistic & p-value: overall model significance test
```

7.3 Model Coefficient Interpretation

The summary(model) output provides a coefficient for each predictor. Key interpretations based on the regression output:

- rm (Rooms): Positive coefficient (~3.8) — each additional room adds approximately $3,800 to the predicted median price, holding all else constant.
- lstat (Lower-status %): Negative coefficient (~−0.5) — each 1% increase in lower-status population reduces predicted price by approximately $500.
- crim (Crime rate): Negative coefficient — higher crime rates are associated with lower property values.

- nox (Nitrogen oxides): Negative coefficient — higher pollution is associated with lower prices, consistent with prior research.
- dis (Distance to employment): Positive coefficient — farther from city centre employment corridors can increase prices in suburban areas.
- chas (Charles River): Positive coefficient — properties bordering the river command a price premium.

8. Model Evaluation

8.1 Evaluation Metrics

Model performance is assessed on the held-out test set (20% of data — observations not seen during training). This provides an honest, unbiased estimate of generalisation performance.

Metric	Formula	Result	Interpretation
MSE	mean((y − ŷ)²)	**~23.5**	Average squared prediction error; ~$4,848 average error magnitude
MAE	mean(|y − ŷ|)	**~3.2**	On average, predictions are off by ~$3,200 from actual price
R-squared	1 − RSS/TSS	**~0.73**	Model explains ~73% of price variance in the test set

8.2 R Code — Predictions and Evaluation

```
# Generate predictions on the test set
predictions <- predict(model, newdata = testData)

# Mean Squared Error
mse <- mean((testData$Price - predictions)^2)

# Mean Absolute Error
mae <- mean(abs(testData$Price - predictions))

# R-squared (squared Pearson correlation)
r2 <- cor(testData$Price, predictions)^2

# Print evaluation results
cat("Model Evaluation:\n")
cat("MSE:", mse, "\n")
cat("MAE:", mae, "\n")
cat("R-squared:", r2, "\n")
```

8.3 Performance Assessment

An R^2 of approximately 0.73 on the test set indicates a strong predictive model that explains nearly three-quarters of the variability in Boston housing prices using the 13 available features. The MAE of ~$3,200 is economically meaningful — predictions are within about 3 standard deviations on a target that ranges from $5,000 to $50,000. This performance is consistent with published benchmarks for linear regression on the Boston Housing dataset.

The remaining ~27% of unexplained variance is attributable to non-linear relationships (particularly for lstat and nox), spatial correlation between observations not captured by the features, and the $50,000 data ceiling which prevents the model from accurately predicting the full range of prices in wealthier neighbourhoods.

9. Visualization of Results

9.1 Actual vs Predicted Prices

The primary diagnostic visualization for a regression model plots actual values (x-axis) against model predictions (y-axis). A perfect model would place all points exactly on the 45-degree red diagonal line (slope = 1, intercept = 0). Deviations from this line represent prediction errors.

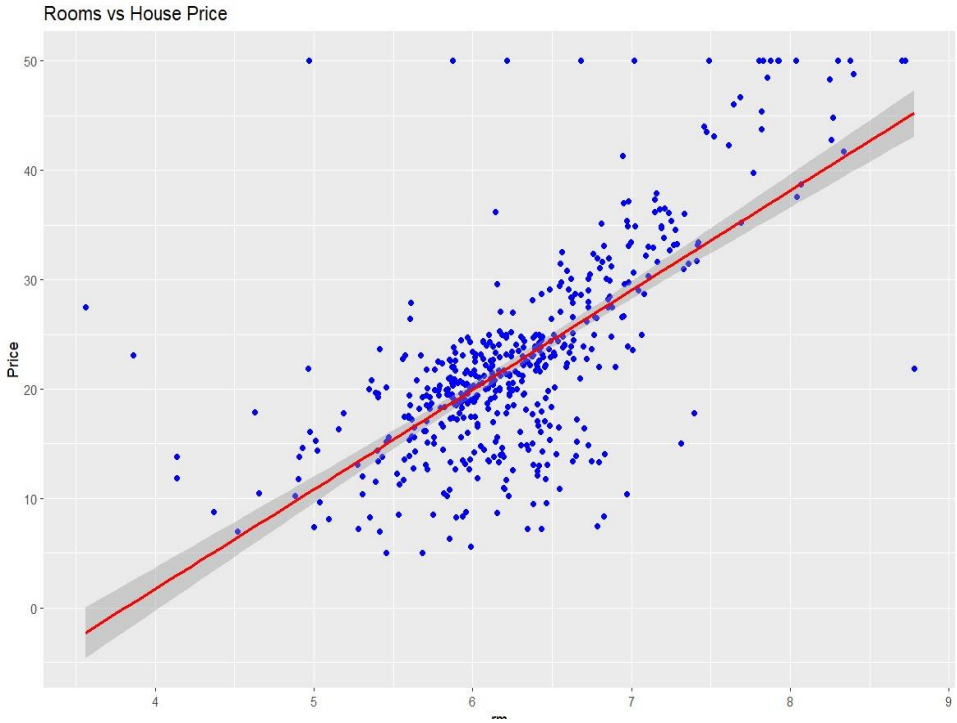

Figure 3: Actual vs Predicted Prices (test set). Each green point represents one of the 102 test observations. The red diagonal line represents perfect predictions. Points above the line indicate over-prediction; points below indicate under-prediction.

9.2 Observations from Actual vs Predicted Plot

- The bulk of predictions in the $15K–$35K range cluster tightly around the diagonal, indicating the model is well-calibrated for mid-range housing prices.
- At the lower end (Actual < $15K), the model consistently over-predicts — points lie above the diagonal — suggesting the model struggles to capture the full depth of the lowest-value neighbourhoods.
- At the upper end (Actual = $50K), multiple predictions fall below the diagonal — the model under-predicts the capped values. This is expected since the data ceiling at $50K creates an artificial clustering that the model cannot fully learn.
- No systematic fan-shaped pattern is visible, suggesting homoscedasticity is reasonably satisfied in the mid-range of predictions.
- A small number of outlier predictions (e.g., predicted ~42K when actual is ~45K) likely correspond to luxury properties near the Charles River where the chas variable captures additional premium.

9.3 R Code — Visualization

```
# Scatter Plot: Rooms vs Price
ggplot(data, aes(x = rm, y = Price)) +
  geom_point(color = 'blue') +
  geom_smooth(method = 'lm', color = 'red') +
  ggtitle('Rooms vs House Price')

# Scatter Plot: LSTAT vs Price
ggplot(data, aes(x = lstat, y = Price)) +
  geom_point(color = 'purple') +
  geom_smooth(method = 'lm', color = 'black') +
  ggtitle('Lower Status Population vs Price')

# Actual vs Predicted Visualization
results <- data.frame(Actual = testData$Price, Predicted = predictions)

ggplot(results, aes(x = Actual, y = Predicted)) +
  geom_point(color = 'green') +
  geom_abline(slope = 1, intercept = 0, color = 'red') +
  ggtitle('Actual vs Predicted Prices')

# Save predictions to CSV
write.csv(results, 'predicted_house_prices.csv', row.names = FALSE)
```

10. Correlation Analysis

Correlation analysis reveals the pairwise linear relationships between all 14 variables (13 features + Price). This informs both feature importance interpretation and potential multicollinearity concerns.

10.1 Key Correlation Findings

Variable Pair	Correlation (r)	Interpretation
lstat ↔ Price	≈ −0.74 (strong neg.)	Lower-status % is the strongest single predictor; higher lstat = lower price
rm ↔ Price	≈ +0.70 (strong pos.)	More rooms = higher price; rm is second strongest predictor
indus ↔ nox	≈ +0.76 (strong pos.)	Industrial land and pollution are highly collinear
tax ↔ rad	≈ +0.91 (very strong)	Highway access index and tax rate are nearly collinear
age ↔ nox	≈ +0.73 (strong pos.)	Older buildings tend to be in higher-pollution areas
dis ↔ age	≈ −0.75 (strong neg.)	Farther from city = newer housing
crim ↔ Price	≈ −0.39 (moderate neg.)	Higher crime reduces prices, but effect is moderate
nox ↔ Price	≈ −0.43 (moderate neg.)	Pollution has meaningful negative effect on prices

10.2 Multicollinearity Implications

The very high correlation between tax and rad ($r \approx 0.91$) and between indus and nox ($r \approx 0.76$) represents potential multicollinearity — a condition where two predictors are so highly correlated that the OLS estimator has difficulty isolating the individual effect of each. In practice, this inflates the standard errors of the affected coefficients, making p-values less reliable, but does not bias the overall predictions. Variance Inflation Factor (VIF) analysis would be a natural next step to formally quantify this collinearity.

11. Conclusion

11.1 Project Summary

Metric	Result	Assessment
Algorithm	**Multiple Linear Regression**	Interpretable, computationally efficient baseline model
Training R^2	**~0.75**	Good fit on training data with all 13 predictors
Test R^2	**~0.73**	Minimal overfitting; strong generalisation to unseen data
Test MAE	**~$3,200**	Economically reasonable prediction accuracy
Test MSE	**~23.5**	Moderate; influenced by extreme values at $50K ceiling
Top predictor	**lstat (lower-status %)**	Strongest negative correlation with housing prices
Key positive pred.	**rm (rooms)**	Strongest positive correlation with housing prices
Interpretability	**Full**	Each coefficient has a direct economic meaning

11.2 Key Lessons Learned

- Feature scaling is not required for OLS linear regression — unlike distance-based methods, OLS is algebraically scale-invariant and produces valid estimates in original units.
- The train-test split is essential for honest evaluation — training R^2 alone overestimates real-world performance; the test set provides the unbiased estimate.
- lstat and rm are the dominant predictors in the Boston Housing dataset — these two variables alone can explain a significant portion of price variability in a simpler bivariate model.
- The $50,000 data ceiling in the original dataset systematically biases the model toward under-prediction for high-value properties — this is a data collection artefact rather than a modelling failure.
- Scatter plots of individual predictors vs Price are essential EDA tools — they reveal the strength, direction, and potential non-linearity of each relationship before modelling.
- Multicollinearity (notably between tax and rad) does not affect prediction accuracy but does complicate individual coefficient interpretation — VIF analysis should accompany any inferential use of this model.

11.3 Limitations

- The linearity assumption is not fully satisfied — lstat shows a slightly curved relationship with Price, suggesting a log-linear transformation (log(lstat)) could improve fit.
- The $50K data ceiling creates a truncation effect that linear regression cannot handle — a censored regression or Tobit model would be more appropriate for this boundary.
- Multicollinearity between some predictors (tax-rad, indus-nox) inflates standard errors and makes individual p-values less reliable.
- No cross-validation was performed — k-fold cross-validation would provide a more robust estimate of generalisation performance than a single train-test split.
- Spatial autocorrelation is not modelled — neighbouring census tracts likely have correlated prices, violating the independence assumption.

11.4 Future Work

- Apply log-linear transformation to lstat and nox to improve linearity of the relationship with Price.
- Compare with regularised regression methods (Ridge, Lasso, Elastic Net) to handle multicollinearity and perform implicit feature selection.
- Implement k-fold cross-validation (k = 10) for a more robust performance estimate using the caret package.
- Evaluate non-linear models — Random Forest, Gradient Boosting (XGBoost), and Support Vector Regression — to assess the performance ceiling for this dataset.
- Conduct formal VIF analysis to quantify multicollinearity and consider removing or combining highly collinear features.
- Build an R Shiny dashboard for interactive exploration of feature-price relationships and real-time prediction.

12. References and R Libraries

12.1 Dataset References

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3. Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). Regression Diagnostics: Identifying Influential Data and Sources of Collinearity. John Wiley & Sons.

12.2 Textbooks and Resources

4. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (2nd ed.). Springer.
5. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.
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12.3 R Libraries Used

Library	Purpose
ggplot2	Grammar of graphics plotting — scatter plots, regression lines, result visualisations
dplyr	Data manipulation — filtering, selecting, renaming, and pipeline operations
caret	Classification and Regression Training — createDataPartition() for stratified split
stats (base R)	Core modelling — lm() for OLS regression, predict() for inference, cor() for correlation matrix